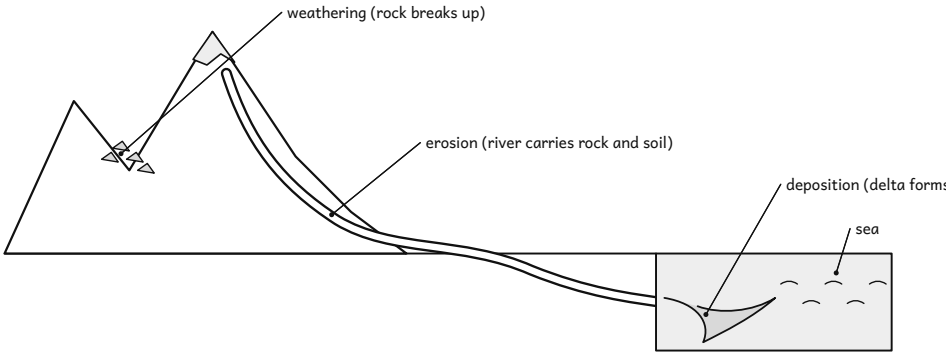
EARTH AND SPACE

# Earth's Changing Surface

Weathering, erosion and deposition, patterns in nature, and sudden changes such as floods, cyclones and earthquakes




**In this pack**

**Learn it:** How the Earth's surface changes ★☆☆
**Fact check:** Shaping the land ★☆☆
**Think it through:** Causes and patterns ★★☆
**Investigate:** Do plants stop soil erosion? ★★☆
**Real-world case study:** Ready for the storm: saving lives from cyclones ★★★
**Review and challenge:** Show what you know ★★★

**This pack belongs to:** _____ **Class:** _____

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LEARN IT · B3 · CHANGING SURFACE

# How the Earth's surface changes

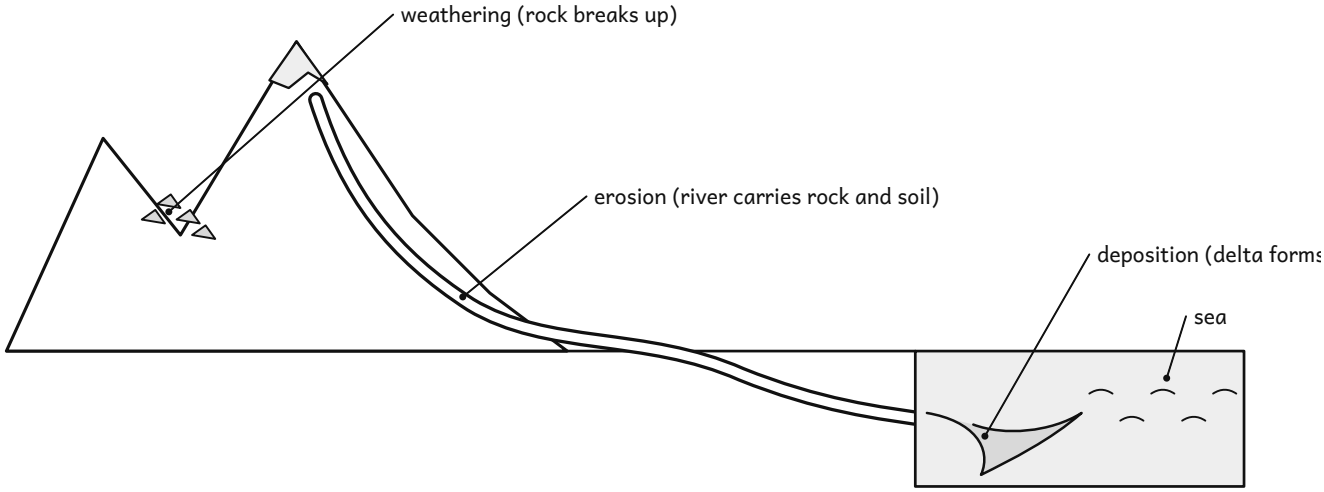
★☆☆
Grade 5

Name: Class: Date:

**Learning goals**

- Explain weathering, erosion and deposition, and how water shapes the land.
- Use evidence to find patterns in natural processes.
- Describe how sudden events and extreme weather change the Earth's surface.

Mountains look as if they will last for ever, and rivers seem to stay in the same place. But the Earth's surface is always changing. Some changes are so **slow** that we cannot see them in a lifetime; others happen **suddenly**, in minutes.




*From mountain to sea: weathering, erosion and deposition.*

### Weathering: rocks break up

Rain, ice, heat and cold, and plant roots slowly crack and break rocks into smaller pieces. Water that freezes in a crack expands and splits the rock. This breaking-up is called **weathering**.

### Erosion: rocks and soil move

Moving water, wind and ice carry the broken pieces away. This is **erosion**. Rivers are the strongest force of erosion in Bangladesh. A fast river in flood can carry huge amounts of sand and silt and can wash away its banks.

### Deposition: new land forms

When a river slows down, especially where it meets the sea, it drops the sand and silt it was carrying. This is **deposition**. Over thousands of years the deposits build up into flat, fertile land called a **delta**.

**Bangladesh connection**

Bangladesh sits on the **Bengal delta**, the largest river delta in the world, built by the Ganges, Brahmaputra and Meghna rivers. New islands called **chars** appear in rivers, and old riverbanks are eaten away. Every year riverbank erosion destroys homes and farmland, especially along the Jamuna and Padma.

### Patterns in nature

Scientists look for **patterns**. In Bangladesh, rivers flood most in the **monsoon** (June to September). Cyclones in the Bay of Bengal are most common in **April–May** and **October–November**. Knowing these patterns helps people prepare.

### Sudden changes

- **Earthquakes:** the Earth's outer layer is broken into giant **tectonic plates** that move a few centimetres a year. When they suddenly slip, the ground shakes. Bangladesh lies close to where three plates meet.
- **Landslides:** heavy rain soaks hillsides until soil and rock slide down, as has happened in the Chattogram hill districts.
- **Cyclones:** powerful storms with very strong winds and a **storm surge**, a wall of sea water pushed onto the coast.
- **Floods:** can drop new silt (good for farming) but also destroy crops, roads and homes.

### People change the land too

Cutting down forests on hills makes landslides more likely because roots no longer hold the soil. Taking sand from rivers can make banks collapse. Planting trees and grass protects soil from erosion.

**Key words**

| | |
|---|---|
| **weathering** | rocks breaking into smaller pieces |
| **erosion** | rock and soil being carried away by water, wind or ice |
| **deposition** | dropping of sand and silt in a new place |
| **delta** | flat land built up where a river meets the sea |
| **tectonic plate** | a giant piece of the Earth's outer layer |
| **storm surge** | sea water pushed onto land by a cyclone |
| **char** | a new island formed from river deposits |

# Shaping the land



Name: Class: Date:

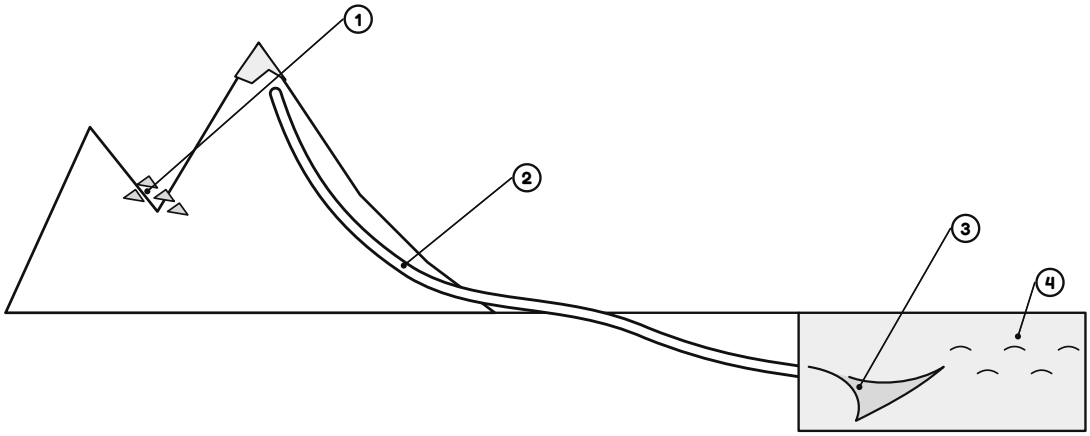
**Learning goals**

- Label the processes.
- Sort fast and slow changes.

## 1 Label the diagram. [4]

LABELS

**deposition (delta forms)** **weathering (rock breaks up)** **sea**
**erosion (river carries rock and soil)**




1 _____ 2 _____

3 _____ 4 _____

## 2 Match each word to its meaning. [5]

Draw a line to match each pair, or write the letter.

| | |
|---|---|
| weathering ● | ● **A** sand and silt being dropped |
| erosion ● | ● **B** land built where a river meets the sea |
| deposition ● | ● **C** sea water pushed onto the land by a cyclone |
| delta ● | ● **D** rock and soil being carried away |
| storm surge ● | ● **E** rocks breaking into pieces |

**3** **Sort the changes into slow and sudden.** [6]

SORT THESE

**rock weathering in a cliff** **a delta growing** **an earthquake** **a cyclone storm surge** **a river valley getting deeper** **a landslide**

| Slow changes | Sudden changes |
|---|---|
| | |
| | |
| | |

**4** **Fill in the gaps.** [3]

WORD BANK

**monsoon** **winter** **desert** **tectonic** **delta**

Bangladesh is built on the world's largest _____. Rivers flood most during the _____. The Earth's outer layer is made of _____ plates.

# Causes and patterns

★★☆
Grade 5

Name: Class: Date:

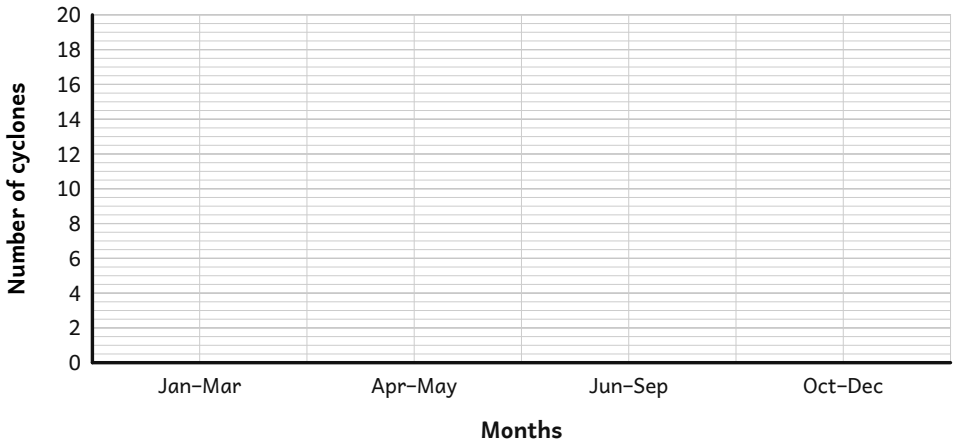
**Learning goals**

- Explain causes of change.
- Use a pattern to make predictions.

**1** **Explain how a river can both destroy land and make new land.** [2]

**2** **This table shows (example) the number of cyclones that formed in the Bay of Bengal in each season over 20 years.** [5]

| Months | Jan–Mar | Apr–May | Jun–Sep | Oct–Dec |
|---|---|---|---|---|
| Cyclones | 1 | 12 | 4 | 19 |




**a)** Draw the bar chart.

**b)** Describe the pattern.

**c)** In which months should coastal people be most prepared?

**3** **Why are landslides more likely on hills where the trees have been cut down?** [2]

**4** **Floods are dangerous, but farmers in Bangladesh also say "the flood brings gold". What do they mean?** [2]

INVESTIGATE · B3 · CHANGING SURFACE

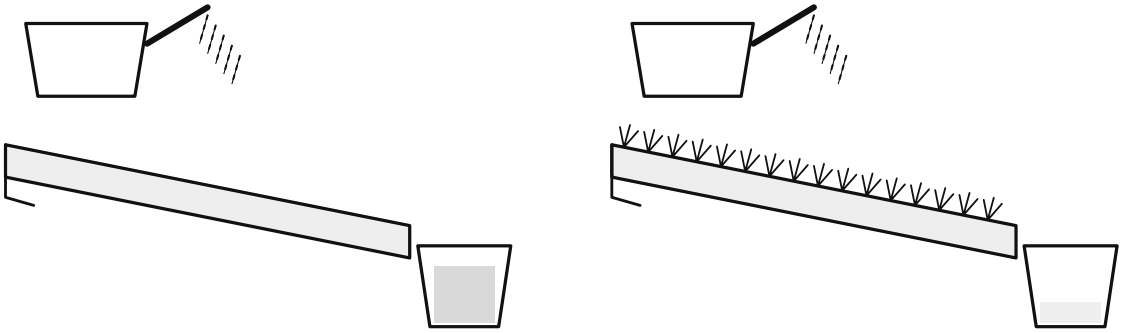
# Do plants stop soil erosion?

★★☆
Grade 5

Name: Class: Date:

**Learning goals**

- Use a model to investigate erosion.
- Control variables.
- Draw conclusions and evaluate the model.



**bare soil** **soil with grass**

*Two trays at the same slope. The same amount of water is poured on each.*

Class 5D tested two trays: one of bare soil and one where grass had grown for three weeks. They poured 500 ml of water from the same height onto each tray and weighed the soil washed into the cup. They repeated it three times.

**1** **Complete the table and find the averages.** [2]

| Tray | Test 1 (g) | Test 2 (g) | Test 3 (g) | Average (g) |
|---|---|---|---|---|
| bare soil | 62 | 58 | 60 | |
| soil with grass | 9 | 12 | 9 | |

**2** **List three things they kept the same.** [3]

.....

.....

**3** **Write a conclusion.** [2]

.....

.....

**4** **Explain your conclusion using what roots and leaves do.** [2]

.....

.....

**5** **How is this model like *a* real hillside in the Chattogram hills, *a*nd how is it different?** [2]

# Ready for the storm: saving lives from cyclones

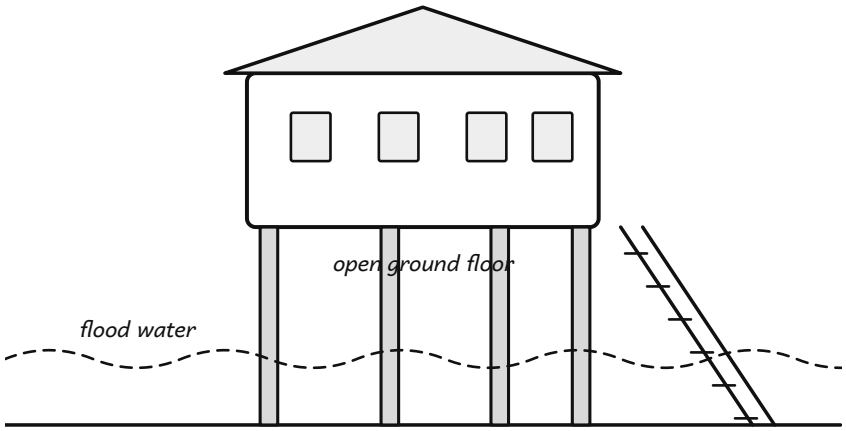
★★★ Grade 5

**Name:** **Class:** **Date:**

**Learning goals**

- Use data to show change over time.
- Explain how science and people reduce risk.
- Suggest further actions.

Science | Geography | History | Maths | Citizenship




*A cyclone shelter on raised pillars, used as a school in normal times.*

In November **1970** the Bhola cyclone struck the coast of what is now Bangladesh. With almost no warning and few safe buildings, an estimated **300 000 to 500 000** people died, mostly drowned by the storm surge.

After this disaster, the **Cyclone Preparedness Programme** began in 1972. Today tens of thousands of trained volunteers warn villages using radios, megaphones and flags, and help people reach thousands of raised **cyclone shelters**. Weather scientists use satellites to track storms days before they arrive.

When Cyclone **Sidr** hit in 2007, about **3400** people died: still a terrible loss, but far fewer than in 1970. When Cyclone **Amphan** hit in 2020, more than 2 million people were moved to shelters and fewer than 30 people died in Bangladesh. The mangroves of the Sundarbans also helped to slow the wind and waves.

Sources: Bangladesh Cyclone Preparedness Programme; UN and Red Cross reports. Death tolls are estimates.

**1** **Complete the table.** [3]

| Cyclone | Year | Approximate deaths in Bangladesh |
|---|---|---|
| Bhola | 1970 | |
| Sidr | 2007 | |
| Amphan | 2020 | |

**2** **Describe the pattern in the deaths over time and give three reasons for it.** [4]

**3** **Why are cyclone shelters built on tall pillars with an open ground floor?** [2]

**4** **Some families do not want to leave their homes, animals and belongings. Suggest how volunteers could persuade them, and how shelters could be improved.** [2]

# Show what you know

★★★ Grade 5

Name: Class: Date:

**Learning goals**

- Use key vocabulary.
- Answer test-style questions.

**1** **Find the key words.**

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| G | O | J | V | T | R | W | J | W | A | B |
| F | N | N | C | C | Y | C | L | O | N | E |
| L | K | R | H | E | R | O | S | I | O | N |
| O | K | U | I | Y | E | O | W | T | S | L |
| O | T | P | B | V | P | J | M | D | T | S |
| D | V | D | J | J | E | L | S | R | J | V |
| R | A | E | I | M | K | R | A | N | W | I |
| O | L | L | T | D | T | O | S | T | U | P |
| C | C | T | I | P | F | P | W | I | E | T |
| K | N | A | M | F | O | W | P | O | L | E |
| U | E | B | C | H | A | R | T | C | S | T |

**char cyclone delta erosion flood plate river rock silt**

**2** **Which process drops sand and silt to form a delta?** [1]

- ◯ weathering
- ◯ erosion
- ◯ deposition
- ◯ evaporation

**3** **What causes earthquakes?** [1]

- ◯ heavy rain
- ◯ tectonic plates suddenly moving
- ◯ strong wind
- ◯ the Moon

**4** **Sort these into helpful and harmful effects of a flood.** [5]

SORT THESE

**new silt for fields** **fills ponds with fish** **homes washed away** **crops destroyed** **dirty water spreads disease**

| Helpful | Harmful |
|---|---|
| | |
| | |
| | |

**5** **Challenge. Why do you think Bangladesh has more new land (chars) in some places and loses land in others, even in the same year?** [2]

.....

.....

.....

**How am I doing? Colour or tick a circle.**

| I can... | Not yet | Nearly | Got it! |
|---|---|---|---|
| explain weathering, erosion and deposition | ○ | ○ | ○ |
| explain how rivers build the Bengal delta | ○ | ○ | ○ |
| find a pattern in data about natural events | ○ | ○ | ○ |
| describe earthquakes, landslides, cyclones and floods | ○ | ○ | ○ |
| use a model to show how plants reduce erosion | ○ | ○ | ○ |
| explain how Bangladesh prepares for cyclones | ○ | ○ | ○ |